

1. If a system S is memoryless, it must be _____.
 A. Stable
 B. Causal
 C. Time-invariant
 D. None of the statements is correct.
2. Let $x(t)$ be a non-constant continuous-time signal, and let $y(t) = x(2t)$. Which statements are true? _____.
 A. If $x(t)$ is periodic, then $y(t)$ is periodic.
 B. If $y(t)$ is periodic, then $x(t)$ is periodic.
 C. If $x(t)$ is periodic with period T ; $y(t)$ is periodic with period $2T$.
 D. If $y(t)$ is periodic with period $2T$; $x(t)$ is periodic with period T .
3. The value of $\int_3^{+\infty} (t^3 + 2)\delta(t + 4)dt$ is _____.
 A. -62
 B. 2
 C. 0
 D. 27
4. For a CT harmonically related signal set with periodic exponentials $\{\phi_k(t) = e^{jk\omega_0 t}, k = \pm 1, \pm 2, \dots\}$, $T_0 = \frac{2\pi}{\omega_0}$ is the common period of all exponentials in this set. This statement is _____.
 A. Correct
 B. Wrong
 C. Dependent on the value of k
 D. Dependent on the value of t
5. A CT LTI system S includes two LTI systems (system A and system B) in cascade. The unit impulse responses are $h_1(t)$ and $h_2(t)$, respectively, for system A and system B. What is the unit impulse response of system S ? _____.
 A. $h_1(t) + h_2(t)$
 B. $h_1(t) \times h_2(t)$
 C. convolution of $h_1(t)$ and $h_2(t)$
 D. depends on the input signal

6. $x_{\text{even}}(t)$ is the even part of a real CT signal $x(t)$, and $x_{\text{odd}}(t)$ is the odd part of the real CT signal $x(t)$. $X(j\omega)$ is the Fourier transform of real signal $x(t)$. The CTFT of $x_{\text{even}}(t)$ is _____, and the CTFT of $x_{\text{odd}}(t)$ is _____.
- A. the real part of $X(j\omega)$ B. the magnitude part of $X(j\omega)$
 C. $j \times$ the imaginary part of $X(j\omega)$ D. the phase part of $X(j\omega)$
7. For a given CT signal $f(t) = \cos(\pi t) + \cos(3\pi t) + \cos(5\pi t)$, its fundamental frequency is _____.
 A. 0.5Hz B. 1.5Hz C. 2Hz D. 2.5Hz
8. Which of the following statements are true? _____
- A. Signal's time-shift only affects the amplitude spectrum.
 B. Knowing the frequency response $H(j\omega)$ of a system, the impulse response $h(t)$ of the system can be uniquely determined.
 C. The spectrum of a periodic CT signal is discrete and non-periodic.
 D. An odd function plus DC still contains sinusoidal components in its Fourier series.
9. The Fourier transform of the signal $f(t)$ is $F(j\omega)$, then the Fourier transform of the signal $f(3-t)$ is _____.
 A. $e^{j3\omega} F(-j\omega)$ B. $e^{-j3\omega} F(-j\omega)$ C. $e^{j3\omega} F(j\omega)$ D. $e^{-j3\omega} F(j\omega)$
10. Which of the following function could be the eigen-function of any LTI system? _____
- A. unit impulse function
 B. any even function
 C. complex exponential function

- (1) A continuous-time signal $x(t)$ is shown in Figure 1. Sketch and label the signal $x(4t + 2)$ carefully.
- (2) Given a CT system $y(t) = x(t - 2) + x(2 - t)$ with input $x(t)$ and output $y(t)$, determine whether this CT system is (a) linear or nonlinear, and (b) time-invariant or time-varying. Please provide a detailed proof process.

Let $x(t) = u(t - 3) - u(t - 5)$ and $h(t) = e^{-3t}u(t)$.

- (1) Compute $y(t) = x(t) * h(t)$.
- (2) Compute $g(t) = (\frac{dx(t)}{dt}) * h(t)$.
- (3) How is $g(t)$ related to $y(t)$?

$$\frac{d}{dt}y(t) + 4y(t) = x(t)$$

find the Fourier series representation of $y(t)$ with the following two inputs:

- (1) $x(t) = \cos(2\pi t)$
- (2) $x(t) = \sin(\pi t) + \cos(6\pi t + \frac{\pi}{4})$

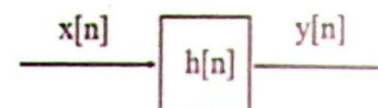
Consider a DT LTI system with unit impulse response $h[n] = (0.2)^n u[n]$, and with an periodic input

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) + \sin\left(\frac{4\pi n}{N}\right)$$

Example 3.17

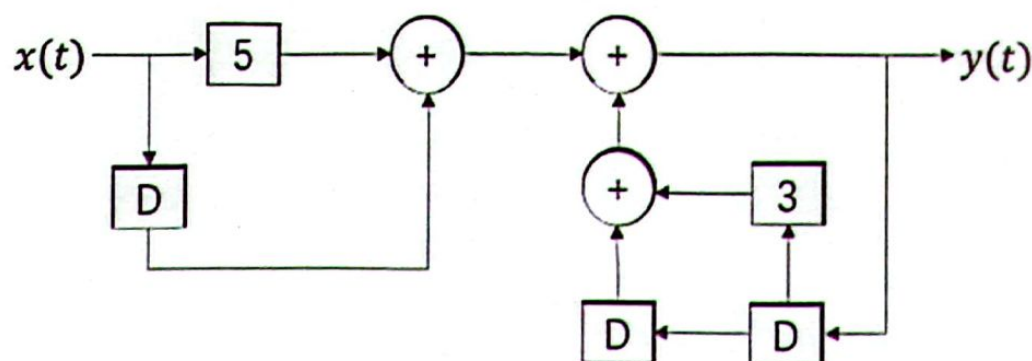
$$h[n] = \alpha^n u[n] \quad |\alpha| < 1$$

$$x[n] = \cos\left(\frac{2\pi n}{N}\right) = \frac{1}{2} e^{j\left(\frac{2\pi}{N}\right)n} + \frac{1}{2} e^{-j\left(\frac{2\pi}{N}\right)n}$$



- (1) Write the input signal $x[n]$ in the form of Fourier series.
- (2) What is the frequency response of this system?
- (3) Write system output $y[n]$ in the form of Fourier series.

- (1) Determine the differential equation relating the input signal $x(t)$ and the output signal $y(t)$.
- (2) Determine the frequency response $H(j\omega)$ of this LTI system.



$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{j\omega + 5}{-\omega^2 + j7\omega + 12}$$

- (1) Determine a differential equation relating the input signal $x(t)$ and the output signal $y(t)$ of S.
- (2) Determine the unit impulse response $h(t)$ of S.
- (3) What is the output signal of S when the input signal is $x(t) = e^{-t}u(t)$?